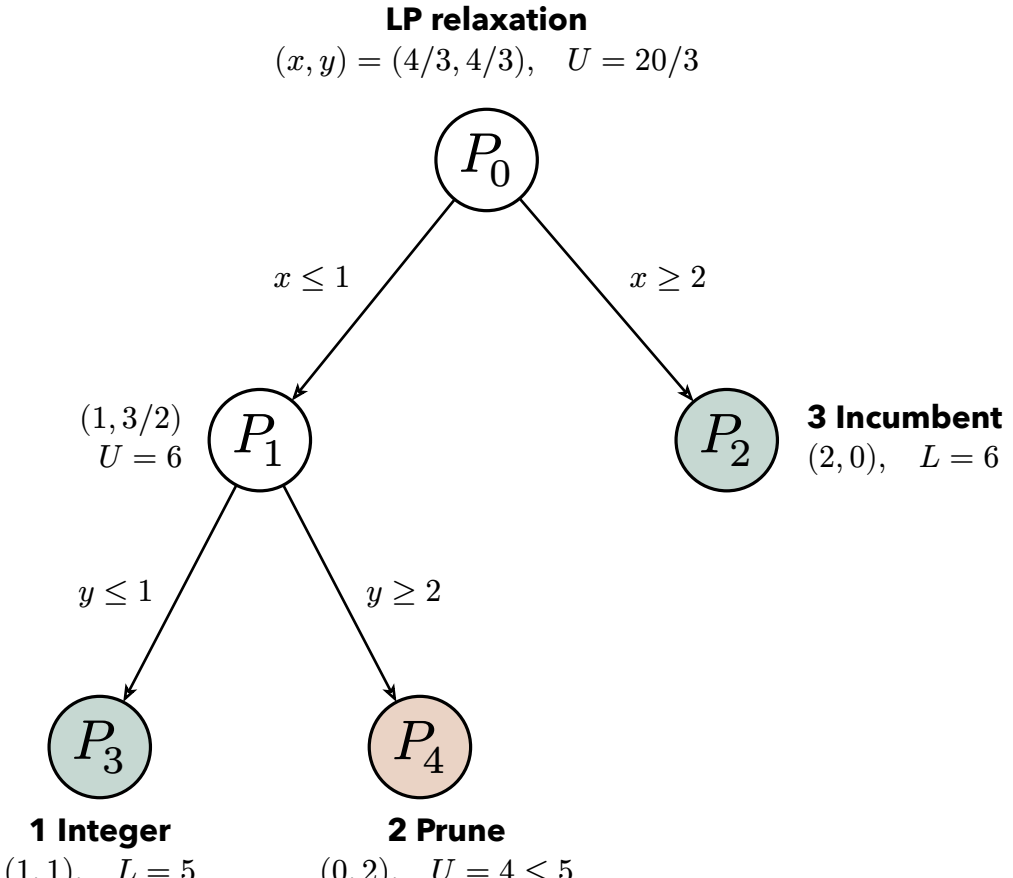
Objective $\max z = 3x + 2y$
Constraints $2x + y \le 4, \quad x + 2y \le 4, \quad x, y \in \mathbb{Z}_{\ge 0}$



Optimal: $(x, y) = (2, 0), \quad z^* = 6$

U = upper bound from the linear-programming (LP) relaxation
L = best integer objective found so far (the incumbent)